

Representation of Groups

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Chapter 1

Representation of Groups and Group Algebras

1.1 Basic Definitions

Suppose now G is a finite group and k is a field.

1. A **k -linear representation** of G is a group homomorphism $\rho : G \rightarrow (V)$ where V is a k -vector space. The space V is called the **representation space** of ρ and $\dim_k V$ (if finite) is called the **degree** of the representation.
2. Let ρ and ρ' be two representations of the same group G in vector spaces V and V' . The **transforms** of V and V' is a k -linear map from V to V' such that $\tau \circ \rho(s) = \rho'(s) \circ \tau$ for all $s \in G$. These representations are said to be **similar** (or isomorphic) if τ is an isomorphism.

A k -linear representation $\rho : G \rightarrow (V)$ is equivalent to a left $k[G]$ -module V . In this case, the transforms between two representations are exactly the $k[G]$ -module homomorphisms.

Let $\rho : G \rightarrow (V)$ be a k -linear representation of a finite group G in the vector space V .

1. A subspace W of V is called a **subrepresentation** if it is a $k[G]$ -submodule of V .
2. A representation is called **irreducible** if it has no proper non-zero subrepresentation.

Let V_i be k -linear representation of a finite group G .

1. The **sum representation** $V_1 \oplus V_2$ is defined on the vector space $V_1 \oplus V_2$ by $g(v_1, v_2) = (gv_1, gv_2)$ for all $g \in G, v_1 \in V_1, v_2 \in V_2$. (the direct sum of $k[G]$ -modules $V_1 \oplus V_2$).
2. The **tensor product representation** $V_1 \otimes V_2$ is defined on the vector space $V_1 \otimes V_2$ by $g(v_1 \otimes v_2) = (gv_1) \otimes (gv_2)$ for all $g \in G, v_1 \in V_1, v_2 \in V_2$. (inner tensor product of $k[G]$ module $V_1 V_2$)
3. The n -th tensor power $V^{\otimes n}$ is again a representation of G by this rule, and the exterior powers $\Lambda^n(V)$ and symmetric powers $Sym^n(V)$ are subrepresentations of it.
4. The dual $V^* = Hom_k(V, k)$ of V is also a representation of G by the rule $\langle g \cdot f, \rangle = \langle f, g^{-1} \rangle$

For representations V, U, W of G , we have $V \otimes (U \oplus W) = (V \otimes U) \oplus (V \otimes W)$,
 $\Lambda^k(V \oplus W) = \bigoplus_{a+b=k} \Lambda^a V \otimes \Lambda^b W$,
 $\Lambda^k(V^*) = k(V^*) = \Lambda^k(V)^*$,

1.2 Maschke's Theorem

In this section, we always assume that k be a field whose characteristic does not divide the order of the finite group G . Let V_i be $k[G]$ -module for $i = 1, 2$ and $f : V_1 \rightarrow V_2$ be a k -linear map, then $F := \frac{1}{|G| \sum_{g \in G} \rho_g^2 \cdot f \cdot (\rho_g^1)^{-1}}$ is a $k[G]$ -module homomorphism from V_1 to V_2 , where $1/|G|$ is the inverse of the order of G in the field k .

[Maschke] If the characteristic of the field k does not divide the $|G|$, then $k[G]$ is semisimple. Let V be a $k[G]$ -module, W be a submodule of V . And let W' be a k -complement of W in V , and let p be the corresponding projection of V onto W . Define a map $P : V \rightarrow W$ by $P(v) := \frac{1}{|G| \sum_{g \in G} g \cdot p(g^{-1}v)}$. Then P is a $k[G]$ -projection from V to W (i.e. P is a $k[G]$ -module homomorphism with $P^2 = P$ and $W = Im(P)$), so $\ker(P)$ is a $k[G]$ -complement of W in V .

It follows from $|G| \leq \infty$ that $k[G]$ is a semi-simple Artinian algebra. Thus the regular representation $R_G := {}_{k[G]}k[G]$ has a unique decomposition $R_G \cong$

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$\bigoplus_{i=1}^m V_i^{\oplus n_i}$ where V_i are all irreducible representations of G non-isomorphic to each other, and n_i is the multiplicity of V_i in R_G . ($n_i = \#_{k[G]}(R_G, V_i)$)

Every representation of G is a direct sum of V_i 's.(that is why we only need to study irreducible representations of finite degree)

The dimension of an irreducible representation of G divides the order of G .

Chapter 2

Character Theory

In this chapter, we will study the character theory of finite degree representations over an algebraically closed field (especially C).

2.1 Characters of Representations

Let G be a group and V be a $k[G]$ -module of finite dimension (or say V is a representation of G).

1. The **character** χ of the representation is defined by $\chi(s) := \text{Tr}(\rho_s)$.
2. If W be a submodule of V , the **subcharacter** χ_W of W is defined by $\chi_W(s) := \text{Tr}(\rho_s|_W)$.
3. A character χ is called **irreducible** if it is the character of an irreducible representation.

If χ is the character of a representation V of degree n , we have:

1. $\chi(1) = n$,
2. $\chi(s^{-1}) = \overline{\chi(s)}$ for $s \in G$,
3. $\chi(tst^{-1}) = \chi(s)$ for $s, t \in G$. $\chi(ab) = \chi(ba)$

Let V_1, V_2 be two linear representations of G . Then:

1. $\chi_{V_1 \oplus V_2} = \chi_{V_1} + \chi_{V_2}$.

2. $\chi_{V_1 \otimes V_2} = \chi_{V_1} \cdot \chi_{V_2}$.
3. $\chi_{\wedge^2 V}(g) = \frac{1}{2} (\chi_V(g)^2 - \chi(g^2))$.
4. $\chi_{\text{Sym}^2 V}(g) = \frac{1}{2} (\chi_V(g)^2 + \chi(g^2))$.

2.2 Schur's lemma and

Let f, g be k -valued functions on a finite group G . The **inner product** of f and g is defined by $(f, g) := \frac{1}{|G|} \sum_{s \in G} f(s)g(s^{-1})$.

[Schur's lemma] Let k be a algebraically closed field and G be a finite group. Let $\rho^1 : G \rightarrow (V_1)$ and $\rho^2 : G \rightarrow (V_2)$ be two irreducible representations of G , and let $f : V_1 \rightarrow V_2$ be a transform. Then:

1. If ρ^1 and ρ^2 are not isomorphic, we have $f = 0$.
2. If $\rho^1 \cong \rho^2$ ($\rho^1 = \rho^2$ and $V_1 = V_2$) then f is a homothety (i.e., a scalar multiple of the identity).

Let h be a linear mapping of V_1 into V_2 , and put: $h^0 = \frac{1}{|G|} \sum_{g \in G} (\rho_g^2)^{-1} h \rho_g^1$. Then:

1. If ρ^1 and ρ^2 are not isomorphic, we have $h^0 = 0$. In this case, $(r_{i_2 j_2}, r_{i_1 j_1}) = \frac{1}{|G|} \sum_{g \in G} r_{i_2 j_2}(g^{-1}) r_{i_1 j_1}(g) = 0$ for arbitrary i_1, i_2, j_1, j_2 .
2. If $V_1 = V_2$ and $\rho^1 = \rho^2$, h^0 is a homothety of ratio $(1/n) \text{Tr}(h)$, with $n = \dim(V_1)$. In this case, $(r_{i_2 j_2}, r_{i_1 j_1}) = \frac{1}{|G|} \sum_{g \in G} r_{i_2 j_2}(g^{-1}) r_{i_1 j_1}(g) = \frac{1}{n} \delta_{i_1 i_2} \cdot \delta_{j_1 j_2}$

2.3 Orthogonality Relations

Let V and W be two irreducible representations of G . Then $(\chi_V, \chi_W) = \begin{cases} 1, & \text{if } V \cong W \\ 0, & \text{otherwise} \end{cases}$

Let V be a linear representation of G , with character χ , and suppose V decomposes into a direct sum of irreducible representations: $V = W_1^{\oplus m_1} \oplus \dots \oplus W_h^{\oplus m_h}$

1. $m_k = (\chi, \chi_{W_k})$.
2. $\chi = \sum_{k=1}^h m_k \chi_{W_k}$
3. $(\chi, \chi) = \sum_{k=1}^h m_k^2$

2.3.1 n_i

The character r_G of the regular representation of G is given by $r_G(g) = \begin{cases} |G|, & g = e \\ 0, & g \neq e \end{cases}$

Let the decomposition of the regular representation of G into irreducible components be $R_G \cong \bigoplus_{i=1}^h W_i^{\oplus n_i}$, where W_i are all irreducible representations non-isomorphic to each other. Then:

1. $\dim W_i = (r_G, \chi_i) = n_i$
2. $\sum n_i^2 = |G|$
3. If $g \neq e$, then $r_G(g) = \sum_{i=1}^h n_i \chi_i(g) = 0$

2.3.2 h

Let $f : G \rightarrow k$ be any function on G , and for a representation V of G set $F := \sum_{g \in G} f(g) \cdot \rho_g$. Then F is a $k[G]$ -module homomorphism for all V if and only if f is a class function on G . If $\rho : G \rightarrow (V)$ is an irreducible representation with character χ , then F is a homothety of ratio $\lambda = \frac{1}{n} \sum_{g \in G} f(g) \chi(g) = \frac{|G|}{n} (f, \bar{\chi})$ where $n = \dim V$.

The all characters $\chi_1, \chi_2, \dots, \chi_h$ form an orthonormal basis of the space $\mathcal{H}$ of class functions on G with values in k .

The number of non-isomorphic irreducible representations of G is equal to the number of conjugacy classes of G .

Let $s \in G$, and let $c(s)$ be the number of elements in the conjugacy class of s . Then

1. the class function equal to 1 on the class of s and equal to 0 elsewhere is given by $f_s(t) = \frac{c(s)}{|G|} \sum_{i=1}^h \chi_i(s^{-1}) \chi_i(t)$
2. For $t \in G$ not conjugate to s , we have $\sum_{i=1}^h \chi_i(s) \chi_i(t^{-1}) = \begin{cases} |G|/c(s), & \text{if } t \text{ is conjugate to } s \\ 0, & \text{if } t \text{ is not conjugate to } s \end{cases}$

The character of any irreducible representation of dimension greater than 1 assumes the value 0 on some conjugacy class of the group.

2.3.3 Canonical projection formula

The $k[G]$ -projection p_i of V onto $V_i = W_i^{\oplus n_i}$ is given by the formula: $p_i = \frac{n_i}{|G|} \sum_{g \in G} \chi_i(g^{-1}) \rho_g$

2.4

The dimension of an irreducible representation of G divides the order of G .

Chapter 3

Induced representation

In this chapter, we always assume that G is a finite group and k is a field whose characteristic does not divide the order of G . And assume that all representations are of finite degree.

3.1

Let H be a subgroup of a group G .

1. If (ρ, W) is a representation of H . Define the **induced representation** $Ind_H^G W$ of G by $Ind_H^G W := k[G] \otimes_{k[H]} W$ as left $k[G]$ -module. [Mackey's definition] Let $\{g_i \mid i \in I\}$ be a complete set of representatives of the left cosets of H in G . Then as a k -vector space, $Ind_H^G W \cong \bigoplus_{i \in I} g_i \otimes W$ and the action of G is given by $s \cdot (g_i \otimes w) = (s \cdot g_i) \otimes w = (g_j \cdot h) \otimes w = g_j \otimes (h \cdot w)$ where $sg_i = g_j h$ with $h \in H$.
2. If (π, V) is a representation of G . Define the **restricted representation** $Res_H^G V$ of H by $Res_H^G V := k[H] \otimes_{k[G]} V \quad (\cong_{k[H]} V)$ as left $k[H]$ -module. $\pi|_H : H \rightarrow (V)$

Two functors Ind_H^G and Res_H^G are defined between the category of $k[H]$ -modules and the category of $k[G]$ -modules.

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- 1.
2. $Ind_{G_2}^{G_1} Ind_{G_3}^{G_2} W \cong Ind_{G_3}^{G_1} W$

3.2 Frobenius reciprocity

If f is a class function on H with values in k , consider the $\text{Ind}_H^G f(s) := \frac{1}{|H|} \sum_{g \in G} \sum_{s' \in H} f(g^{-1}sg) f(s')$ called the **induced class function** of f from H to G , denoted by $\text{Ind}_H^G f$. Then:

1. $\text{Ind}_H^G f$ is a class function on G .
2. If χ is the character of a representation of H , then $\text{Ind}_H^G \chi$ is the character of the induced representation $\text{Ind}_H^G W$.

If f is a class function on G with values in k , consider the $\text{Res}_H^G f(s) := f(s)$ called the **restricted class function** of f from G to H , denoted by $\text{Res}_H^G f$. Then:

1. $\text{Res}_H^G f$ is a class function on H .
2. If χ is the character of a representation of G , then $\text{Res}_H^G \chi$ is the character of the restricted representation $\text{Res}_H^G V$.

Let G be a group and χ_i be the character of V_i , $(\chi_2, \chi_2) = (V_1, V_2) := \dim_k {}_{k[G]}(V_1, V_2)$

[Frobenius reciprocity] Given a $k[G]$ -module V , and a $k[H]$ -module W , we have ${}_{k[H]}(W, {}_H^G V) \cong {}_{k[G]}(\text{Ind}_H^G W, V)$ as k -vector spaces.

[Frobenius reciprocity] Let H be a subgroup of a finite group G . If χ is the character of a representation of H and ψ is the character of a representation of G , then $(\text{Ind}_H^G \chi, \psi)_G = (\chi, \text{Res}_H^G \psi)_H$. Thus it is true for class functions. Let W be the representation space of χ and V be the representation space of ψ . By the definition of inner product and the previous lemma, we have $(\text{Ind}_H^G \chi, \psi)_G = \dim_k {}_{k[G]}(\text{Ind}_H^G W, V) = \dim_k {}_{k[H]}(W, \text{Res}_H^G V) = (\chi, \text{Res}_H^G \psi)_H$

3.3 Mackey's Decomposition and irreducibility criterion

Let $G = \bigsqcup_{g \in K \backslash G/H} KgH$ be a double coset decomposition of G .

$k[G] \cong \bigoplus_{g \in K \backslash G/H} k[KgH]$ as $k[K]$ - $k[H]$ -bimodules.

For a fixed $g \in G$, we have $k[KgH] \cong k[K] \otimes_{k[K \cap gHg^{-1}]} k[gH]$ as $k[K]$ - $k[H]$ -bimodules.

[Mackey's Decomposition Theorem] Then $\text{Res}_K^G \circ \text{Ind}_H^G \cong \bigoplus_{g \in K \backslash G/H} \text{Ind}_{K \cap gHg^{-1}}^K \circ {}^g \text{Res}_{H \cap gKg^{-1}}^H$ where ${}^g W$ is the conjugate module of W by g . Given representation W of H , we have $\text{Res}_K^G \circ \text{Ind}_H^G (W) = k[K] \otimes_{k[K]} k[G] \otimes_{k[H]} W$
 $\cong \bigoplus_{g \in K \backslash G/H} k[K] \otimes_{k[K]} k[K] \otimes_{k[K \cap gHg^{-1}]} k[gH] \otimes_{k[H]} W$
 $\cong \bigoplus_{g \in K \backslash G/H} k[K] \otimes_{k[K \cap gHg^{-1}]} k[gH] \otimes_{k[H]} W$

Noted that $k[gH] \otimes_{k[H]} W$ is $k[K \cap gHg^{-1}]$ -module which is isomorphic to ${}^g \text{Res}_{H \cap gKg^{-1}}^H W$ as $k[K \cap gHg^{-1}]$ -module.

Chapter 4

S_3	1	3	2
	1	(12)	(123)
U	1	1	1
U'	1	-1	1
V	2	0	-1

	S_4	1	6	8	6	3
		1	(12)	(123)	(1234)	(12)(34)
$trivialU$		1	1	1	1	1
$alternatingU'$		1	-1	1	-1	1
$standardV$		3	1	0	-1	-1
$V' = V \otimes U'$		3	-1	0	1	-1
$AnotherW$		2	0	-1	0	2

A_4	1	4	4	3
	1	(123)	(132)	(12)(34)
U	1	1	1	1
U'	1	ζ_3	ζ_3^2	1
U''	1	ζ_3^2	ζ_3	1
V	3	0	0	-1

S_5	1	10	20	30	24	15	20
	1	(12)	(123)	(1234)	(12345)	(12)(34)	(12)(345)
U	1	1	1	1	1	1	1
U'	1	-1	1	-1	1	1	-1
V	4	2	1	0	-1	0	-1
V'	4	-2	1	0	-1	0	1
$\wedge^2 V$	6	0	0	0	1	-2	0
W	5	1	-1	-1	0	1	1
W'	5	-1	-1	1	0	1	-1